الجامعة المصرية
للتعلم الإلكتروني الأهلية

EELU
THE EGYPTIAN E-LEARNING UNIVERSITY

Advanced Computer Networks Section 3

What is TCP?

Transmission control Protocol :

- It is a connection-oriented and reliable transport layer protocol (It establishes a connection before data transfer and ensures that data is delivered accurately and in order (3-way handshake)).

- Provides **reliable, ordered, and error-checked delivery** of byte streams.

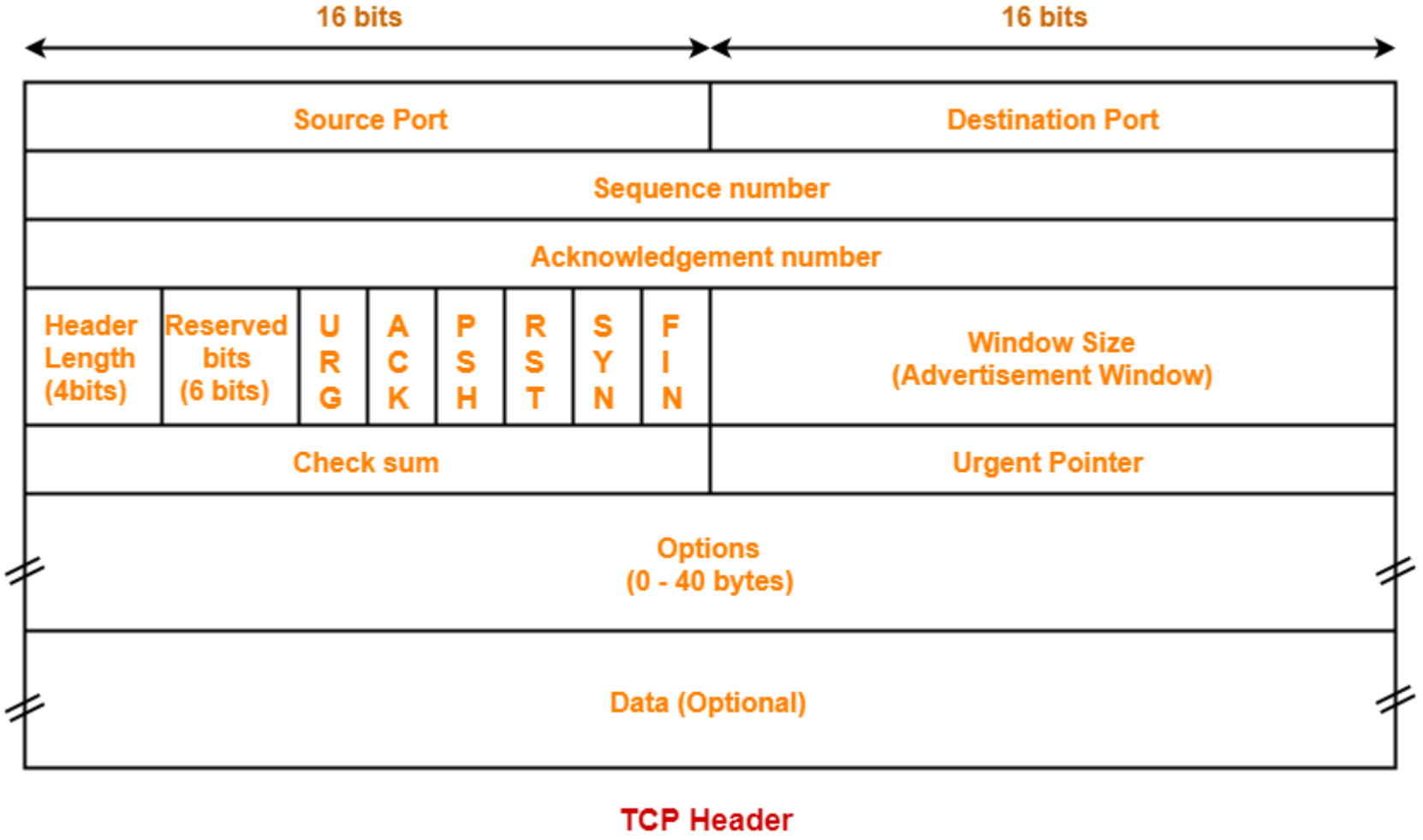
Features:

- **3-Way Handshake:** SYN → SYN/ACK → ACK.
- **Reliability:** retransmission on packet loss.
- **Ordering:** sequence numbers ensure data arrives in order.
- **Acknowledgments (ACKs):** confirm received data.
- **Flow Control:** uses a window size to prevent the sender from overwhelming the receiver.
- **Congestion Control:** uses algorithms like AIMD, Slow Start, Congestion Avoidance.
- **Multiplexing:** uses source and destination ports to manage multiple sessions simultaneously.

When is it used?

Whenever reliability and order matter: web browsing (HTTP/HTTPS), file transfer (FTP), email (SMTP/IMAP), etc.

TCP Header Structure



TCP Header

Main Fields: Source Port, Destination Port, Sequence Number, Acknowledgment Number, Data Offset, Flags (SYN, ACK, FIN, RST, PSH, URG), Window, Checksum, Options.

TCP Header Structure...cont

Field	Size (bytes)	Description
Source Port	2	Port number of the sending application
Destination Port	2	Port number of the receiving application
Sequence Number	4	Identifies the position of the first byte of data in the segment
Acknowledgment Number	4	The next byte expected from the other side
Header Length (Data Offset)	4 bits	Indicates where the data begins
Flags (Control Bits)	6 bits	Include SYN, ACK, FIN, RST, PSH, URG
Window Size	2	Flow control: how much data the receiver can accept
Checksum	2	Error checking for header and data
Urgent Pointer	2	Indicates urgent data if URG flag is set
Options (optional)	Variable	Used for extra features like MSS, window scaling

Minimum header size = 20 bytes.

TCP vs UDP

Feature	TCP	UDP
Connection type	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable
Ordering	Guaranteed	Not guaranteed
Error recovery	Yes	No
Flow control	Yes	No
Congestion control	Yes	No
Speed	Slower	Faster
Header size	20 bytes (min)	8 bytes

How to Capture TCP Traffic in Wireshark

Open Wireshark → Select Interface

Start Capture

Generate TCP Traffic (e.g., In your browser, open an unencrypted website https://gaia.cs.umass.edu or Any website that starts with **http://** (not https://) will make the device establish a **TCP connection on port 80**).

Stop Capture

TCP Connection Establishment (Three-Way Handshake)

1.SYN: Client sends SYN to start a connection)Client → Server(.

2.SYN-ACK: Server replies with SYN + ACK)Server → Client(.

3.ACK: Client sends final ACK → connection established)Client → Server(.

To focus on TCP traffic, apply display filters.

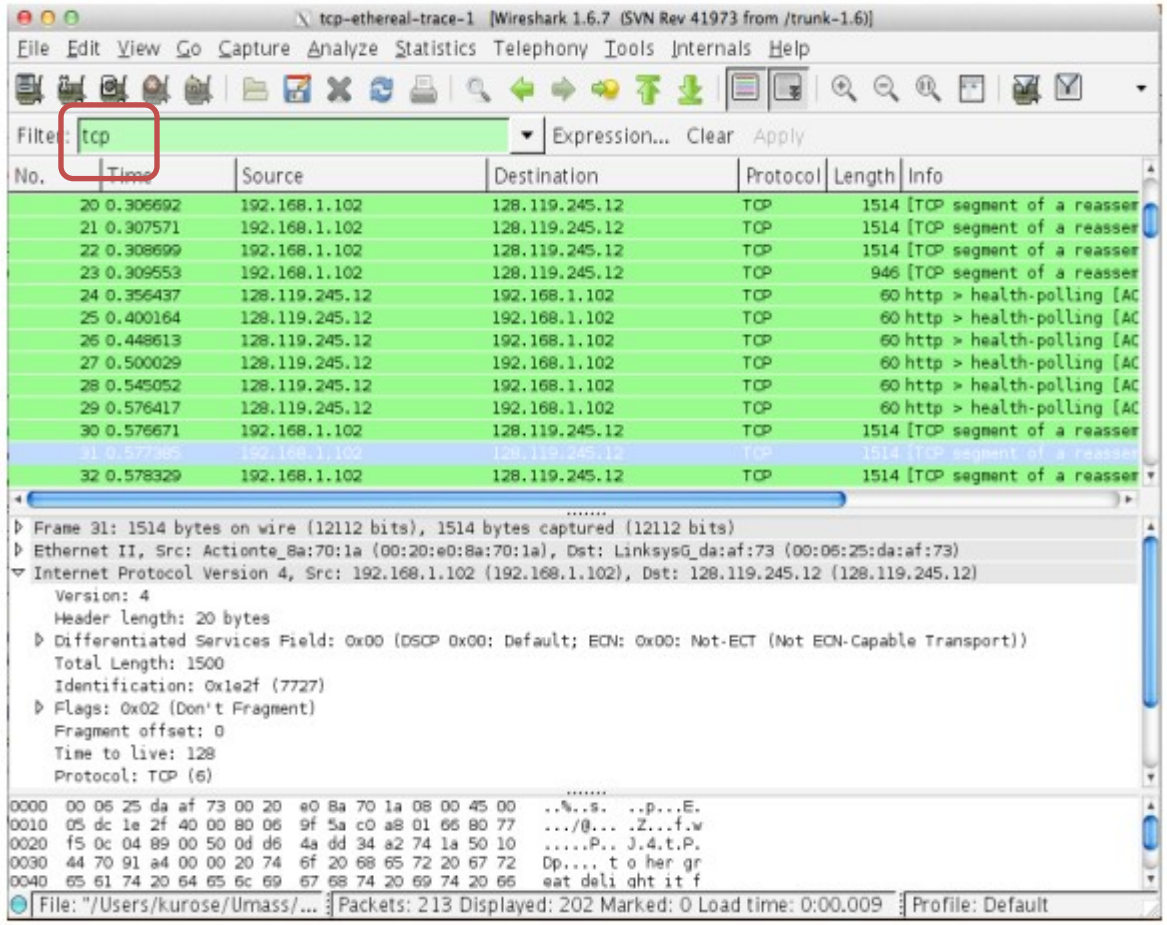
Useful Filters:

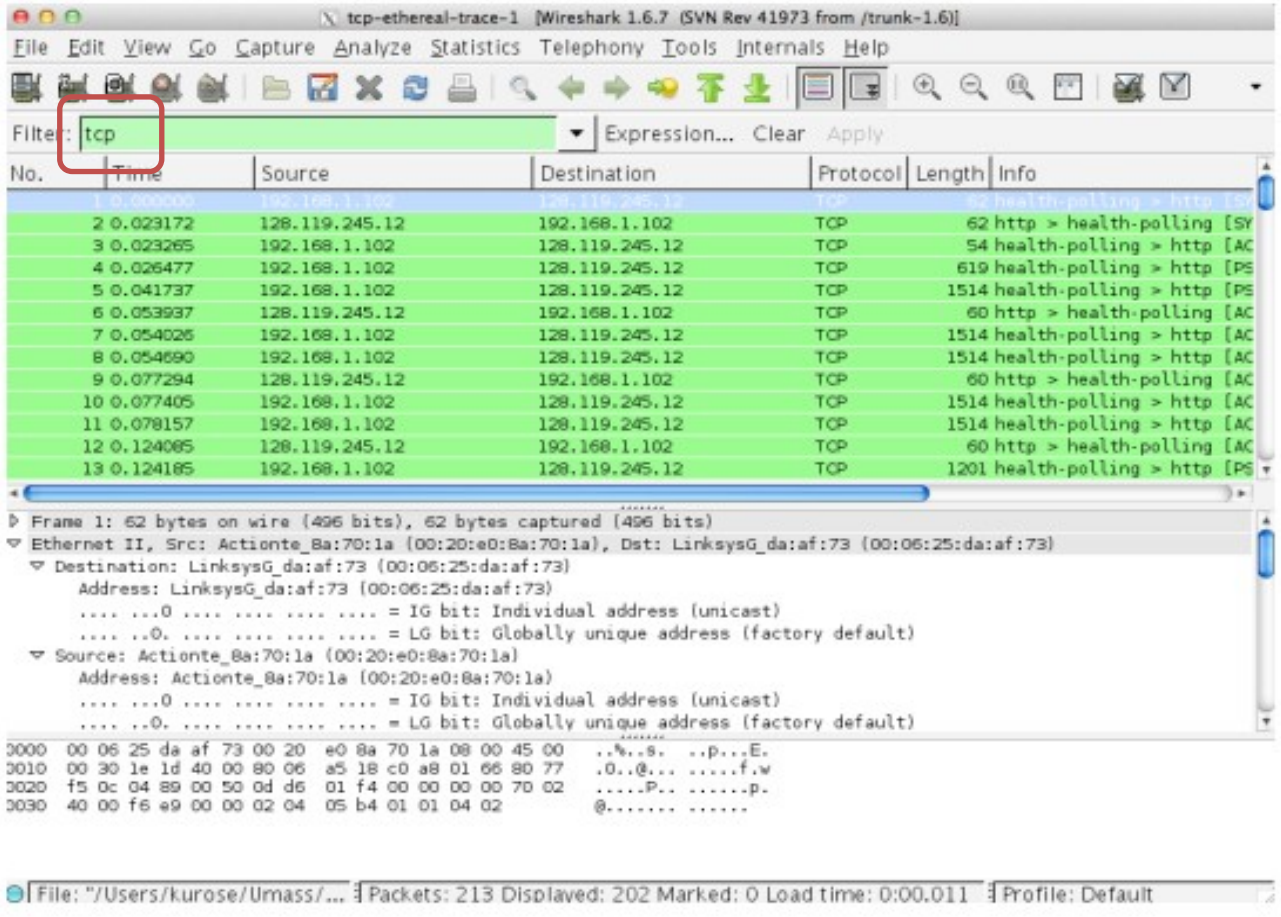
tcp — shows all TCP packets.

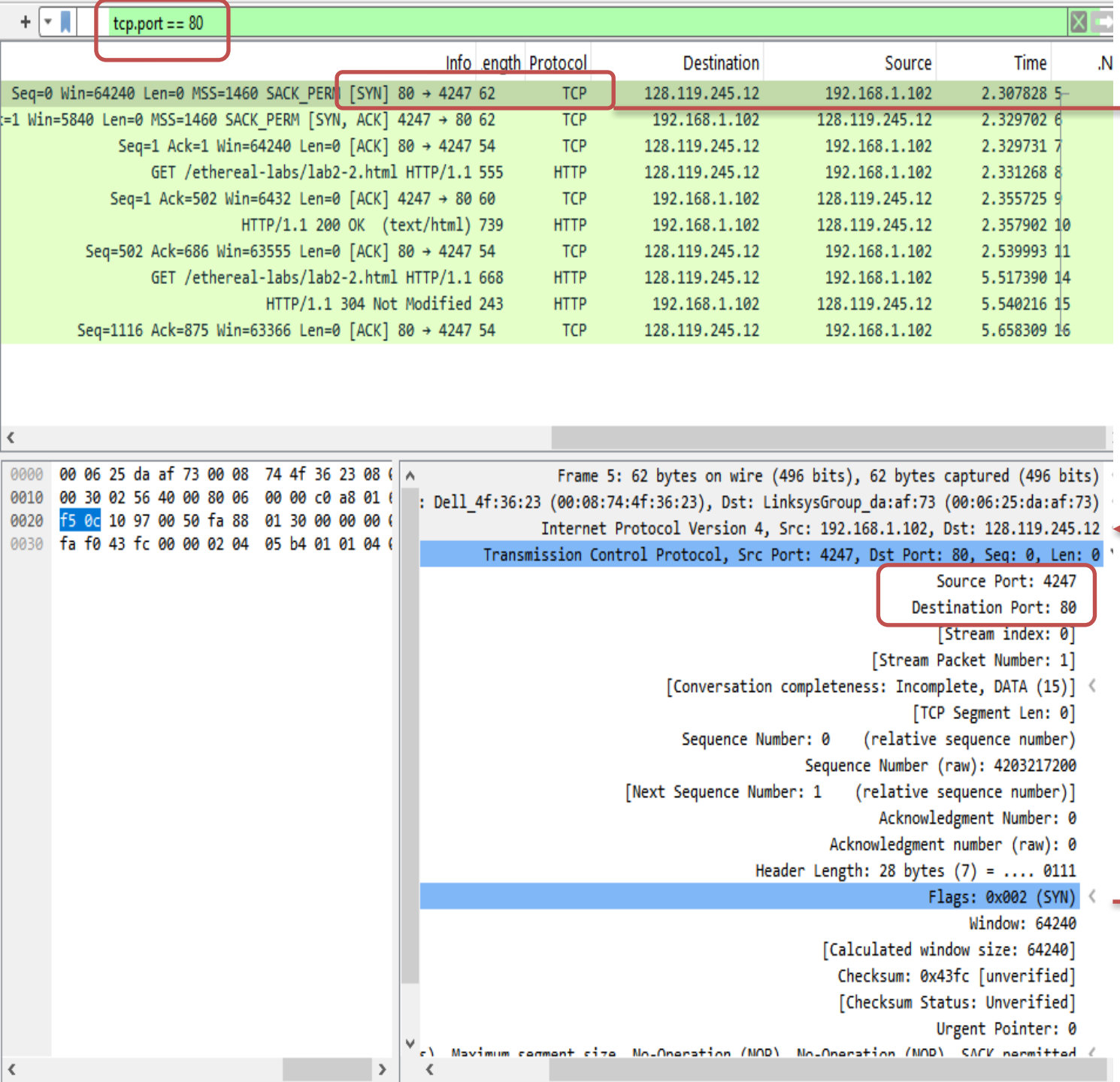
tcp.port == 80 — shows only packets involving port 80) shows HTTP(.

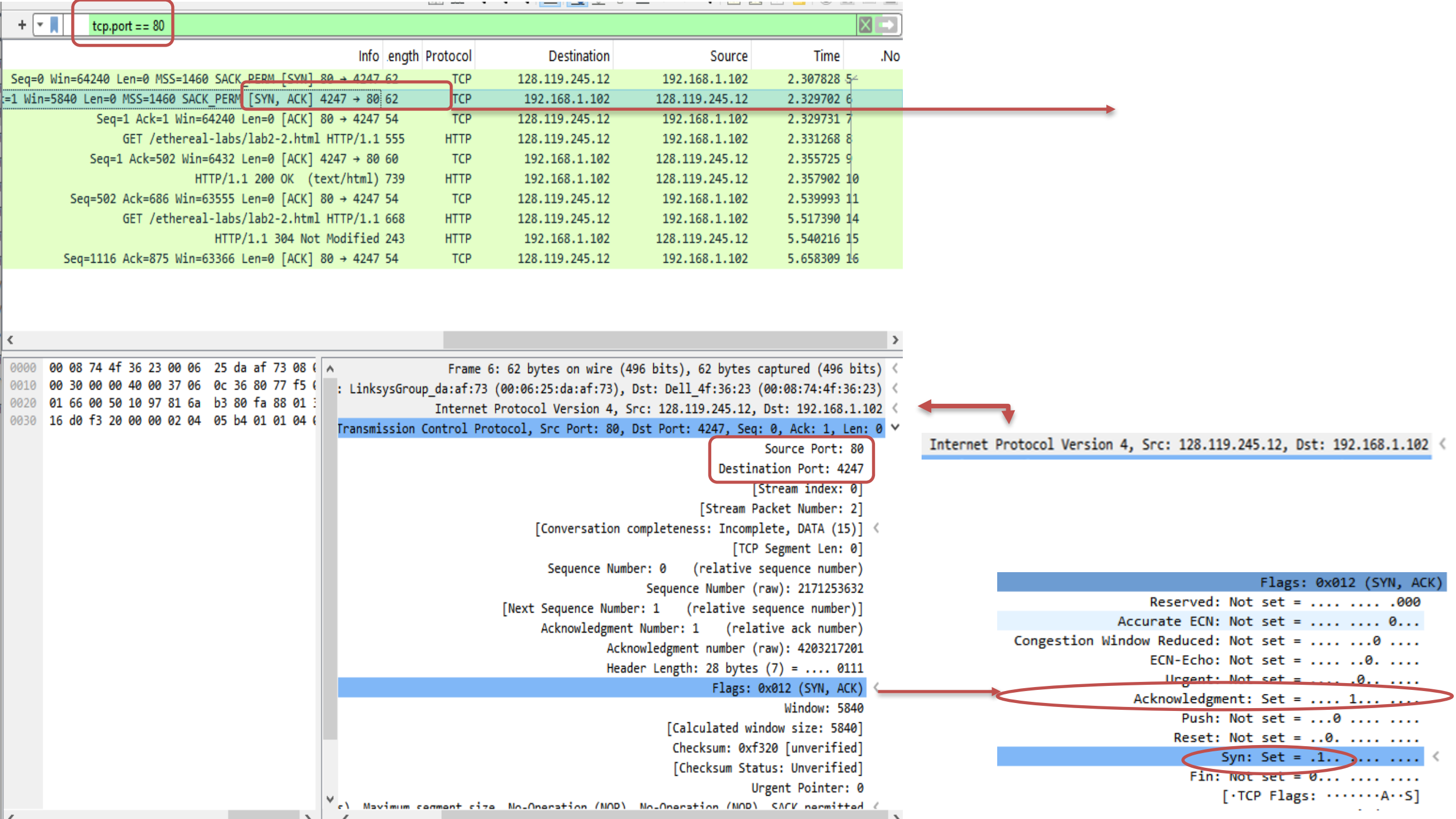
tcp.port == 21 — packets from a specific host using port 21) shows FTP(.

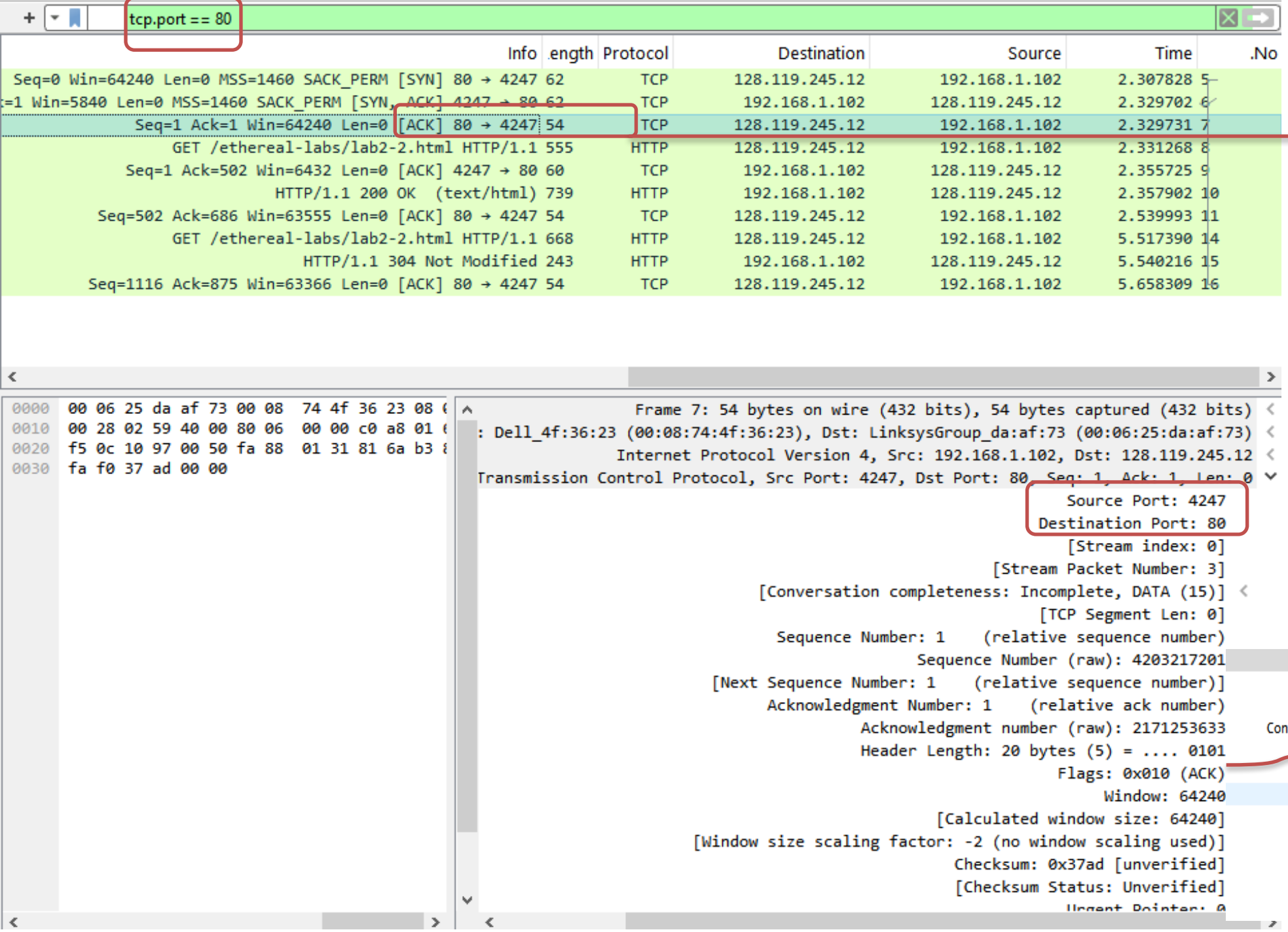
tcp.port == 443 → shows HTTPS.


tcp-ethereal-trace-1 [Wireshark 1.6.7 (SVN Rev 41973 from /trunk-1.6)]
File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help
Filter: tcp
Expression... Clear Apply
No. Time Source Destination Protocol Length Info
20 0.306692 192.168.1.102 128.119.245.12 TCP 1514 [TCP segment of a reasser
21 0.307571 192.168.1.102 128.119.245.12 TCP 1514 [TCP segment of a reasser
22 0.308699 192.168.1.102 128.119.245.12 TCP 1514 [TCP segment of a reasser
23 0.309553 192.168.1.102 128.119.245.12 TCP 946 [TCP segment of a reasser
24 0.356437 128.119.245.12 192.168.1.102 TCP 60 http > health-polling [AC
25 0.400164 128.119.245.12 192.168.1.102 TCP 60 http > health-polling [AC
26 0.448613 128.119.245.12 192.168.1.102 TCP 60 http > health-polling [AC
27 0.500029 128.119.245.12 192.168.1.102 TCP 60 http > health-polling [AC
28 0.545052 128.119.245.12 192.168.1.102 TCP 60 http > health-polling [AC
29 0.576417 128.119.245.12 192.168.1.102 TCP 60 http > health-polling [AC
30 0.576671 192.168.1.102 128.119.245.12 TCP 1514 [TCP segment of a reasser
32 0.578329 192.168.1.102 128.119.245.12 TCP 1514 [TCP segment of a reasser
Frame 31: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits)
Ethernet II, Src: Actionte_8a:70:1a (00:20:e0:8a:70:1a), Dst: LinksysG_da:af:73 (00:06:25:da:af:73)
Internet Protocol Version 4, Src: 192.168.1.102 (192.168.1.102), Dst: 128.119.245.12 (128.119.245.12)
Version: 4
Header length: 20 bytes
Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))
Total Length: 1500
Identification: 0x1e2f (7727)
Flags: 0x02 (Don't Fragment)
Fragment offset: 0
Time to live: 128
Protocol: TCP (6)
0000 00 06 25 da af 73 00 20 e0 8a 70 1a 08 00 45 00 ..%..s. ..p...E.
0010 05 dc 1e 2f 40 00 80 06 9f 5a c0 a8 01 66 80 77 .../@... .Z...f.w
0020 f5 0c 04 89 00 50 0d d6 4a dd 34 a2 74 1a 50 10 .....P.. J.4.t.P.
0030 44 70 91 a4 00 00 20 74 6f 20 68 65 72 20 67 72 Dp.... t o her gr
0040 65 61 74 20 64 65 6c 69 67 68 74 20 69 74 20 66 eat deli ght it f
File: "/Users/kurose/Umass/... Packets: 213 Displayed: 202 Marked: 0 Load time: 0:00.009 Profile: Default


tcp-ethereal-trace-1 [Wireshark 1.6.7 (SVN Rev 41973 from /trunk-1.6)]
File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help
Filter: tcp
Expression... Clear Apply
No. Time Source Destination Protocol Length Info
2 0.023172 128.119.245.12 192.168.1.102 TCP 62 http > health-polling [SY
3 0.023265 192.168.1.102 128.119.245.12 TCP 54 health-polling > http [AC
4 0.026477 192.168.1.102 128.119.245.12 TCP 619 health-polling > http [PS
5 0.041737 192.168.1.102 128.119.245.12 TCP 1514 health-polling > http [PS
6 0.053937 128.119.245.12 192.168.1.102 TCP 60 http > health-polling [AC
7 0.054026 192.168.1.102 128.119.245.12 TCP 1514 health-polling > http [AC
8 0.054690 192.168.1.102 128.119.245.12 TCP 1514 health-polling > http [AC
9 0.077294 128.119.245.12 192.168.1.102 TCP 60 http > health-polling [AC
10 0.077405 192.168.1.102 128.119.245.12 TCP 1514 health-polling > http [AC
11 0.078157 192.168.1.102 128.119.245.12 TCP 1514 health-polling > http [AC
12 0.124065 128.119.245.12 192.168.1.102 TCP 60 http > health-polling [AC
13 0.124185 192.168.1.102 128.119.245.12 TCP 1201 health-polling > http [PS
Frame 1: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)
Ethernet II, Src: Actionte_8a:70:1a (00:20:e0:8a:70:1a), Dst: LinksysG_da:af:73 (00:06:25:da:af:73)
Destination: LinksysG_da:af:73 (00:06:25:da:af:73)
Address: LinksysG_da:af:73 (00:06:25:da:af:73)
....0 .... = IG bit: Individual address (unicast)
.... ..0. .... = LG bit: Globally unique address (factory default)
Source: Actionte_8a:70:1a (00:20:e0:8a:70:1a)
Address: Actionte_8a:70:1a (00:20:e0:8a:70:1a)
....0 .... = IG bit: Individual address (unicast)
.... ..0. .... = LG bit: Globally unique address (factory default)
0000 00 06 25 da af 73 00 20 e0 8a 70 1a 08 00 45 00 ..%..s. ..p...E.
0010 00 30 1e 1d 40 00 80 06 a5 18 c0 a8 01 66 80 77 .0..@... .....f.w
0020 f5 0c 04 89 00 50 0d d6 01 f4 00 00 00 00 70 02 .....P.. ......p.
0030 40 00 f6 e9 00 00 02 04 05 b4 01 01 04 02 @..... ......
File: "/Users/kurose/Umass/... Packets: 213 Displayed: 202 Marked: 0 Load time: 0:00.011 Profile: Default


tcp.port == 80
Info ength Protocol Destination Source Time .N
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM [SYN] 80 → 4247 62 TCP 128.119.245.12 192.168.1.102 2.307828 5
=1 Win=5840 Len=0 MSS=1460 SACK_PERM [SYN, ACK] 4247 → 80 62 TCP 192.168.1.102 128.119.245.12 2.329702 6
Seq=1 Ack=1 Win=64240 Len=0 [ACK] 80 → 4247 54 TCP 128.119.245.12 192.168.1.102 2.329731 7
GET /ethereal-labs/lab2-2.html HTTP/1.1 555 HTTP 128.119.245.12 192.168.1.102 2.331268 8
Seq=1 Ack=502 Win=6432 Len=0 [ACK] 4247 → 80 60 TCP 192.168.1.102 128.119.245.12 2.355725 9
HTTP/1.1 200 OK (text/html) 739 HTTP 192.168.1.102 128.119.245.12 2.357902 10
Seq=502 Ack=686 Win=63555 Len=0 [ACK] 80 → 4247 54 TCP 128.119.245.12 192.168.1.102 2.539993 11
GET /ethereal-labs/lab2-2.html HTTP/1.1 668 HTTP 128.119.245.12 192.168.1.102 5.517390 14
HTTP/1.1 304 Not Modified 243 HTTP 192.168.1.102 128.119.245.12 5.540216 15
Seq=1116 Ack=875 Win=63366 Len=0 [ACK] 80 → 4247 54 TCP 128.119.245.12 192.168.1.102 5.658309 16
0000 00 06 25 da af 73 00 08 74 4f 36 23 08
0010 00 30 02 56 40 00 80 06 00 00 c0 a8 01
0020 f5 0c 10 97 00 50 fa 88 01 30 00 00 00
0030 fa f0 43 fc 00 00 02 04 05 b4 01 01 04
Frame 5: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)
: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: LinksysGroup_da:af:73 (00:06:25:da:af:73)
Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 4247, Dst Port: 80, Seq: 0, Len: 0
Source Port: 4247
Destination Port: 80
[Stream index: 0]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 4203217200
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
Header Length: 28 bytes (7) = ... 0111
Flags: 0x002 (SYN)
Window: 64240
[Calculated window size: 64240]
Checksum: 0x43fc [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.119.245.12
Flags: 0x002 (SYN)
Reserved: Not set = ... .000
Accurate ECN: Not set = ... 0...
Congestion Window Reduced: Not set = ...0 ....
ECN-Echo: Not set = ...0. ....
Urgent: Not set = ...0.. ....
Acknowledgment: Not set = ... 0... ....
Push: Not set = ...0 ....
Reset: Not set = ..0. ....
Syn: Set = .1.. ....
Fin: Not set = 0... ....
[·TCP Flags: ··········S]


tcp.port == 80
Info .ength Protocol Destination Source Time .No
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM [SYN] 80 → 4247 62 TCP 128.119.245.12 192.168.1.102 2.307828 5
=1 Win=5840 Len=0 MSS=1460 SACK_PERM [SYN, ACK] 4247 → 80 62 TCP 192.168.1.102 128.119.245.12 2.329702 6
Seq=1 Ack=1 Win=64240 Len=0 [ACK] 80 → 4247 54 TCP 128.119.245.12 192.168.1.102 2.329731 7
GET /ethereal-labs/lab2-2.html HTTP/1.1 555 HTTP 128.119.245.12 192.168.1.102 2.331268 8
Seq=1 Ack=502 Win=6432 Len=0 [ACK] 4247 → 80 60 TCP 192.168.1.102 128.119.245.12 2.355725 9
HTTP/1.1 200 OK (text/html) 739 HTTP 192.168.1.102 128.119.245.12 2.357902 10
Seq=502 Ack=686 Win=63555 Len=0 [ACK] 80 → 4247 54 TCP 128.119.245.12 192.168.1.102 2.539993 11
GET /ethereal-labs/lab2-2.html HTTP/1.1 668 HTTP 128.119.245.12 192.168.1.102 5.517390 14
HTTP/1.1 304 Not Modified 243 HTTP 192.168.1.102 128.119.245.12 5.540216 15
Seq=1116 Ack=875 Win=63366 Len=0 [ACK] 80 → 4247 54 TCP 128.119.245.12 192.168.1.102 5.658309 16
0000 00 08 74 4f 36 23 00 06 25 da af 73 08
0010 00 30 00 00 40 00 37 06 0c 36 80 77 f5
0020 01 66 00 50 10 97 81 6a b3 80 fa 88 01
0030 16 d0 f3 20 00 00 02 04 05 b4 01 01 04
Frame 6: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)
: LinksysGroup_da:af:73 (00:06:25:da:af:73), Dst: Dell_4f:36:23 (00:08:74:4f:36:23)
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.102
Transmission Control Protocol, Src Port: 80, Dst Port: 4247, Seq: 0, Ack: 1, Len: 0
Source Port: 80
Destination Port: 4247
[Stream index: 0]
[Stream Packet Number: 2]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 2171253632
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 4203217201
Header Length: 28 bytes (7) = ... 0111
Flags: 0x012 (SYN, ACK)
Window: 5840
[Calculated window size: 5840]
Checksum: 0xf320 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.102
Flags: 0x012 (SYN, ACK)
Reserved: Not set = ... .000
Accurate ECN: Not set = ... 0...
Congestion Window Reduced: Not set = ...0 ....
ECN-Echo: Not set = ...0. ....
Urgent: Not set = ...0.. ....
Acknowledgment: Set = ... 1... ....
Push: Not set = ...0 ....
Reset: Not set = ..0. ....
Syn: Set = .1.. ....
Fin: Not set = 0... ....
[·TCP Flags: ·····A··S]


tcp.port == 80
Info .ength Protocol Destination Source Time .No
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM [SYN] 80 → 4247 62 TCP 128.119.245.12 192.168.1.102 2.307828 5
=1 Win=5840 Len=0 MSS=1460 SACK_PERM [SYN, ACK] 4247 → 80 62 TCP 192.168.1.102 128.119.245.12 2.329702 6
Seq=1 Ack=1 Win=64240 Len=0 [ACK] 80 → 4247 54 TCP 128.119.245.12 192.168.1.102 2.329731 7
GET /ethereal-labs/lab2-2.html HTTP/1.1 555 HTTP 128.119.245.12 192.168.1.102 2.331268 8
Seq=1 Ack=502 Win=6432 Len=0 [ACK] 4247 → 80 60 TCP 192.168.1.102 128.119.245.12 2.355725 9
HTTP/1.1 200 OK (text/html) 739 HTTP 192.168.1.102 128.119.245.12 2.357902 10
Seq=502 Ack=686 Win=63555 Len=0 [ACK] 80 → 4247 54 TCP 128.119.245.12 192.168.1.102 2.539993 11
GET /ethereal-labs/lab2-2.html HTTP/1.1 668 HTTP 128.119.245.12 192.168.1.102 5.517390 14
HTTP/1.1 304 Not Modified 243 HTTP 192.168.1.102 128.119.245.12 5.540216 15
Seq=1116 Ack=875 Win=63366 Len=0 [ACK] 80 → 4247 54 TCP 128.119.245.12 192.168.1.102 5.658309 16
0000 00 06 25 da af 73 00 08 74 4f 36 23 08
0010 00 28 02 59 40 00 80 06 00 00 c0 a8 01
0020 f5 0c 10 97 00 50 fa 88 01 31 81 6a b3
0030 fa f0 37 ad 00 00
Frame 7: 54 bytes on wire (432 bits), 54 bytes captured (432 bits)
: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: LinksysGroup_da:af:73 (00:06:25:da:af:73)
Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 4247, Dst Port: 80, Seq: 1, Ack: 1, Len: 0
Source Port: 4247
Destination Port: 80
[Stream index: 0]
[Stream Packet Number: 3]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 4203217201
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2171253633
Header Length: 20 bytes (5) = ... 0101
Flags: 0x010 (ACK)
Window: 64240
[Calculated window size: 64240]
[Window size scaling factor: -2 (no window scaling used)]
Checksum: 0x37ad [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Flags: 0x010 (ACK)
Reserved: Not set = ... .000
Accurate ECN: Not set = ... 0...
Congestion Window Reduced: Not set = ...0 ....
ECN-Echo: Not set = ...0. ....
Urgent: Not set = ...0.. ....
Acknowledgment: Set = ... 1... ....
Push: Not set = ...0 ....
Reset: Not set = ..0. ....
Syn: Not set = .0.. ....
Fin: Not set = 0... ....
[····TCP Flags: ·····A]

Example Trace source & destination for different protocos……..

HTTP Request

Frame 123: 74 bytes on wire
Ethernet II, Src: 00:11:22..., Dst: 66:77:88...
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 93.184.216.34
Transmission Control Protocol, Src Port: 54321, Dst Port: 80, Seq: 1, Ack: 1, Flags: [SYN]

Then after handshake:

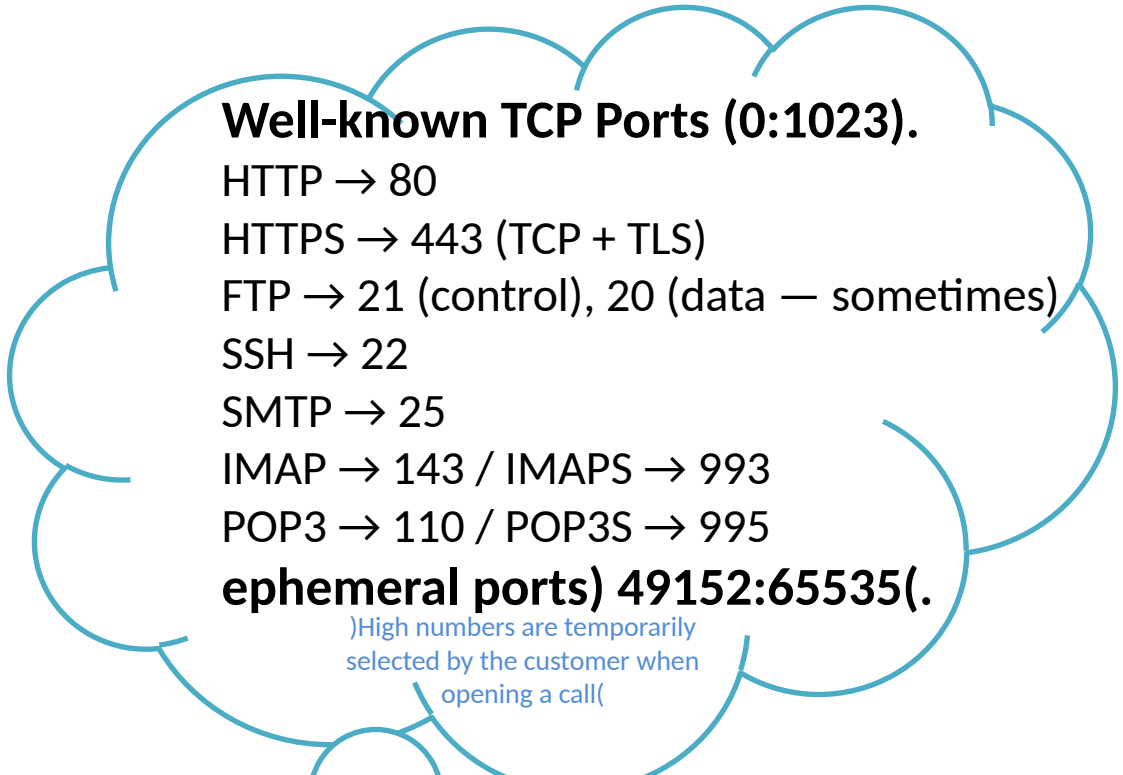
Transmission Control Protocol, Src Port: 54321, Dst Port: 80, Seq: 1, Ack: 1, Flags: [ACK]
Hypertext Transfer Protocol
GET /index.html HTTP/1.1\r\nHost: example.com\r\n...

Explanation: The client uses a random *source port* (54321), and the server listens on *destination port* 80.

FTP Request

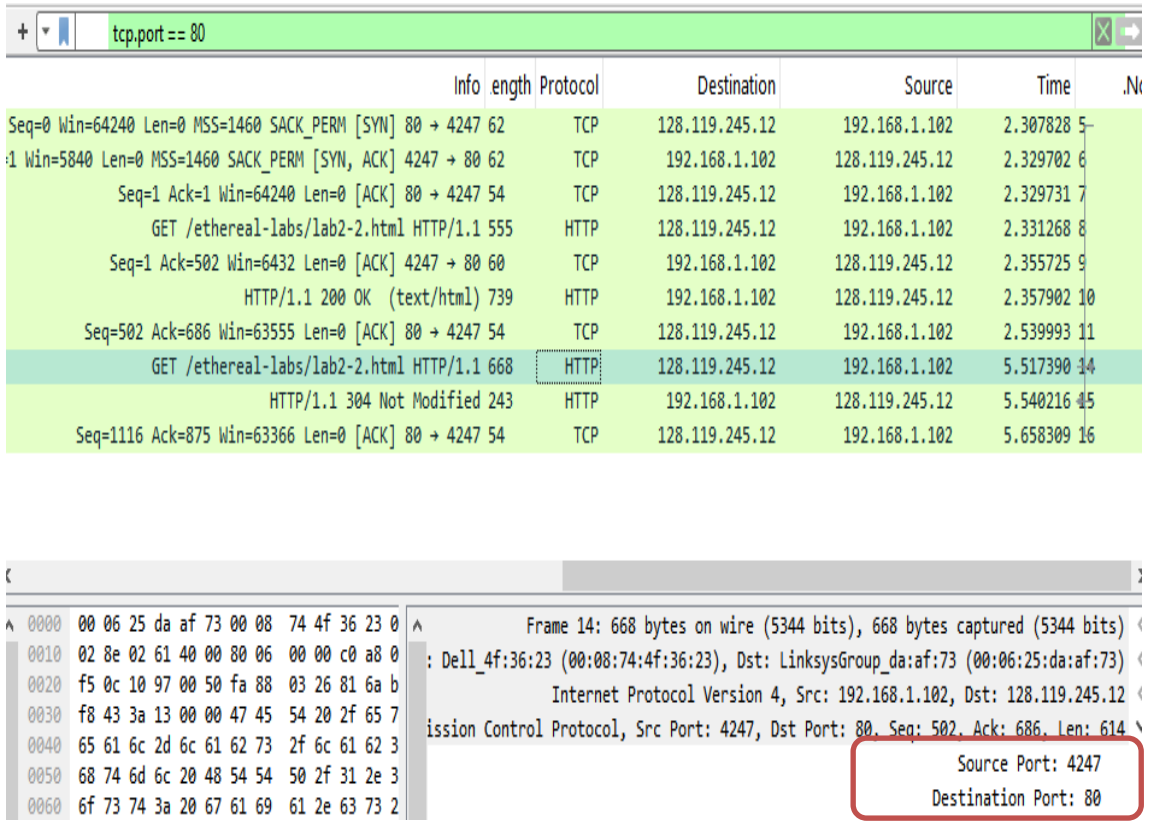
IP Src: 192.168.1.10, Dst: 198.51.100.5
TCP Src Port: 34567, Dst Port: 21, Flags: [PSH, ACK]
Payload: "USER asmaa\r\n"

Explanation: Destination port 21 shows that this is an FTP control connection.

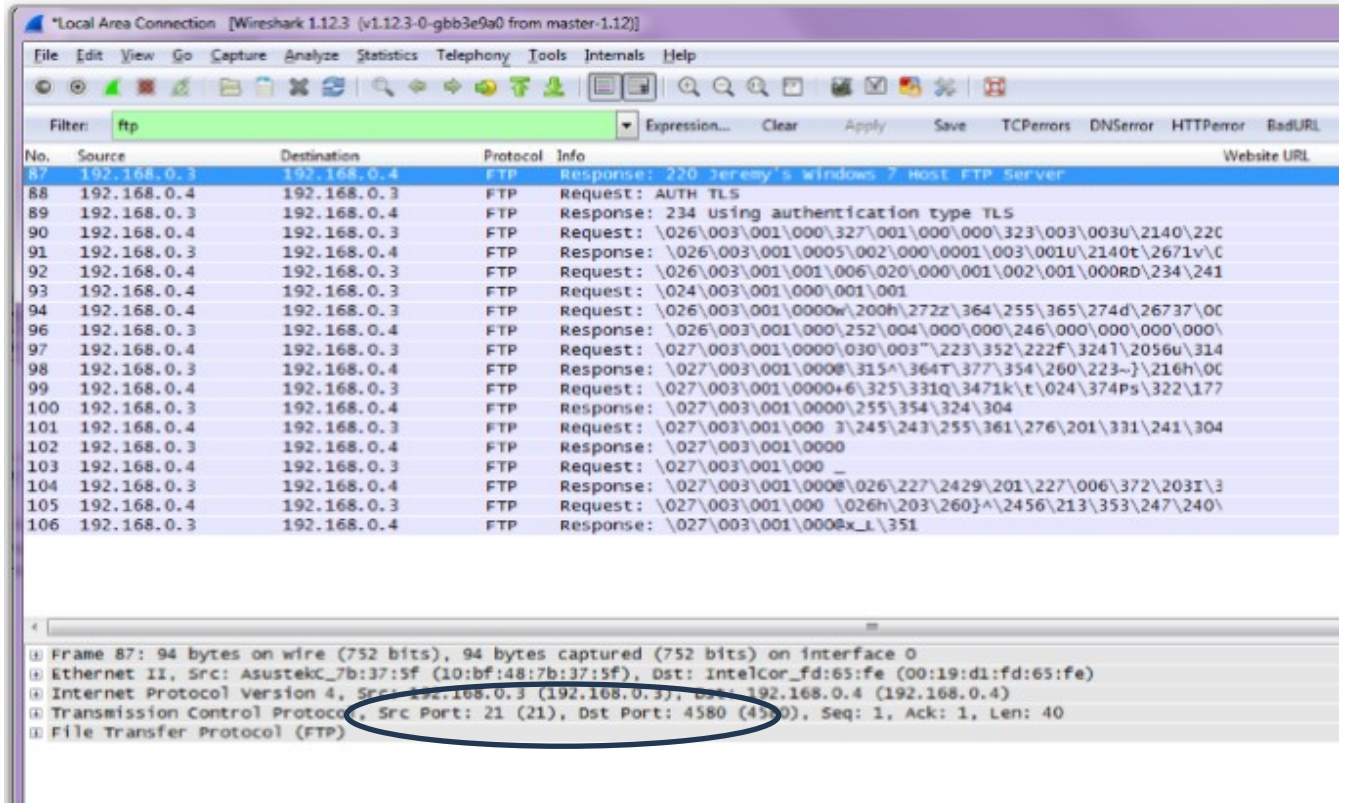


Example Trace source & destination for different protocos……..

HTTP Response



FTP Request



Questions……

From your trace answer the following questions,,,,,,,

Q1:what is the size of the TCP header?

From your trace answer the following questions,,,,,,,

Q2: Which application uses TCP port 80?
Which application uses TCP port 21?

From your trace answer the following questions,,,,,,,

Q3: What is the length of each of the first six TCP segments?

From your trace answer the following questions,,,,,,,

Q4: Capture a TCP communication for a web browsing session (HTTP)

a) Identify the source and destination port numbers in the TCP SYN packet.

b) Which side (client or server) uses the well-known port?

c) What is the typical port number used for HTTP?

Expected answer

Source Port: Random (ephemeral) – e.g., 50432
Destination Port: 80
The **server** uses the well-known port (80).
HTTP typically uses **TCP port 80**.

From your trace answer the following questions,,,,,,,

Q5: Capture an FTP session using Wireshark

a) Identify the source and destination port numbers for the control connection.

b) Which port number is associated with the FTP control channel?

c) Compare the port numbers with those in the HTTP example.

Expected answer

Source Port: Random (ephemeral) – e.g., 49500
Destination Port: **21** (FTP control connection)
FTP uses port **21** for control and port **20** for data transfer.
HTTP used port 80, while FTP uses 21 → both are TCP-based but with different known ports.

From your trace answer the following questions,,,,,,,

Q6: identify the TCP three-way handshake for any protocol you captured (HTTP/FTP).

Label the three packets and specify their flags (SYN, SYN-ACK, ACK).

Expected answer

Packet	Direction	Flags	Description
1	Client → Server	SYN	Request to start connection
2	Server → Client	SYN, ACK	Server acknowledges
3	Client → Server	ACK	Connection established

From your trace answer the following questions,,,,,,,

Q7: identify the TCP header fields visible in Wireshark (Sequence number, Acknowledgment number, Window size).
What do these fields indicate?

Expected answer

- **Sequence number:** Order of bytes sent.
- **Acknowledgment number:** Next byte expected from the other side.
- **Window size:** Flow control (how much data the receiver can handle).

TRY…..

- **Using Wireshark, capture TCP communications for at least two different application protocols (for example, HTTP and FTP).**
  Then answer the following questions based on your capture.

  a) Identify the source and destination port numbers for the TCP SYN packet in each captured protocol.
  What do you observe?

  b) From the same capture, identify the three packets involved in the TCP three-way handshake for one of the protocols (HTTP or FTP).List the flags for each.

  c) Check the TCP header for one packet in your trace.
  Identify the following fields and explain briefly:
  - Sequence Number
  - Acknowledgment Number
  - Window Size
  - Checksum

  Compare TCP and UDP based on your observations from both lab sessions